

Computer Networks

Unit:-1 Data communication Components: Representation of data and its flow Networks , Various Connection Topology, Protocols and Standards, OSI model, Transmission Media, LAN: Wired LAN, Wireless LANs, connecting LAN and Virtual LAN Techniques for Bandwidth utilization: Multiplexing - Frequency division, Time division and Wave division, Concepts on spread spectrum.

Computer Network:-

- ✓ A computer network is a collection of interconnected devices that share resources and information.
- ✓ Computer networks first developed in 1950 for military and defense purpose.
- ✓ The aim of the computer network is the sharing of resources among various devices.

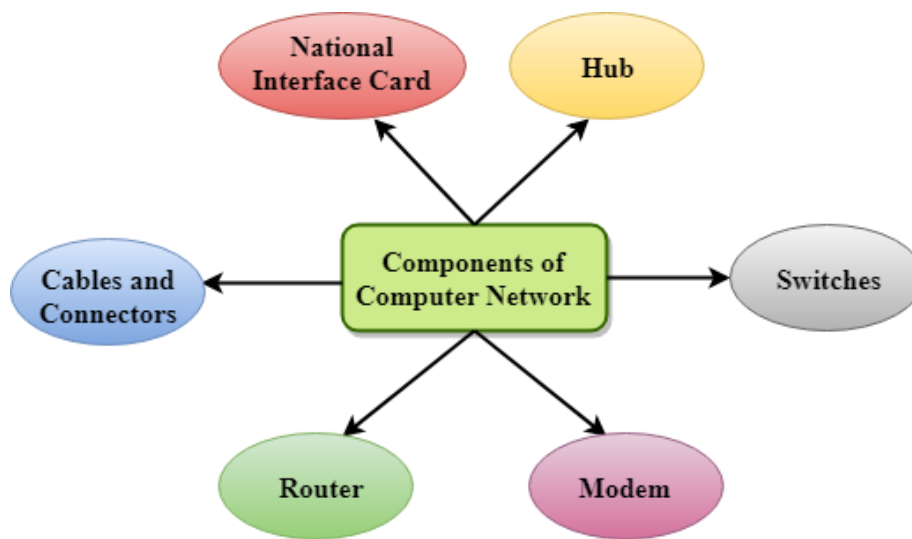
Basic Terminologies of Computer Networks

- **Network:** A network is a collection of computers and devices that are connected together to enable communication and data exchange.
- **Nodes:** Nodes are devices that are connected to a network. These can include computers, Servers, Printers, [Routers](#), [Switches](#), and other devices.
- **Protocol:** A protocol is a set of rules and standards that govern how data is transmitted over a network. Examples of protocols include TCP/IP, [HTTP](#), and [FTP](#).
- **Topology:** Network topology refers to the physical and logical arrangement of nodes on a network. The common network topologies include bus, star, ring, mesh, and tree.
- **Service Provider Networks:** These types of Networks give permission to take Network Capacity and Functionality on lease from the Provider. Service Provider Networks include Wireless Communications, Data Carriers, etc.
- **IP Address:** An IP address is a unique numerical identifier that is assigned to every device on a network. IP addresses are used to identify devices and enable communication between them.
- **DNS:** The Domain Name System (DNS) is a protocol that is used to translate human-readable domain names (such as [www.google.com](#)) into IP addresses that computers can understand.
- **Firewall:** A [firewall](#) is a security device that is used to monitor and control incoming and outgoing network traffic. Firewalls are used to protect networks from unauthorized access and other security threats.

Network Devices:-

- An interconnection of multiple devices, also known as hosts, that are connected using multiple paths for the purpose of sending/receiving data or media.

Components Of Computer Network:



NIC(National interface card):-

- ✓ NIC is a device that helps the computer to communicate with another device.
- ✓ The network interface card contains the hardware addresses, the data-link layer protocol use this address to identify the system on the network so that it transfers the data to the correct destination.

There are two types of NIC: wireless NIC and wired NIC.

- **Wireless NIC:** All the modern laptops use the wireless NIC. In Wireless NIC, a connection is made using the antenna that employs the **radio wave technology**.
- **Wired NIC:** Cables use the **wired NIC** to transfer the data over the medium.

Hub:-

- ✓ Hub is a central device that splits the network connection into multiple devices.
- ✓ When computer requests for information from a computer, it sends the request to the Hub.
- ✓ Hub distributes this request to all the interconnected computers.

Switches:-

- ✓ Switch is a networking device that groups all the devices over the network to transfer the data to another device.
- ✓ A switch is better than Hub as it does not broadcast the message over the network, i.e., it sends the message to the device for which it belongs to.
- ✓ Therefore, we can say that switch sends the message directly from source to the destination.

Cables and connectors:-

Cable is a transmission media that transmits the communication signals. **There are three types of cables:**

- **Twisted pair cable:** It is a high-speed cable that transmits the data over **1Gbps** or more.
- **Coaxial cable:** Coaxial cable resembles like a TV installation cable. Coaxial cable is more expensive than twisted pair cable, but it provides the high data transmission speed.
- **Fibre optic cable:** Fibre optic cable is a high-speed cable that transmits the data using light beams. It provides high data transmission speed as compared to other cables. It is more expensive as compared to other cables, so it is installed at the government level.

Router:-

- ✓ Router is a device that connects the LAN to the internet.
- ✓ The router is mainly used to connect the distinct networks or connect the internet to multiple computers.

Modem:-

- ✓ Modem connects the computer to the internet over the existing telephone line.
- ✓ A modem is not integrated with the computer motherboard.
- ✓ A modem is a separate part on the PC slot found on the motherboard.

Repeater:-

- ✓ A **repeater** in computer networking is a hardware device used to regenerate and amplify network signals to extend the range of a network.
- ✓ It helps in preventing signal degradation caused by distance or interference.

Different Data types in a computer network:

1. Texts
2. Numbers
3. Images
4. Videos
5. Audios

1. Text

- **Representation:** Encoded into binary using standards like:
 - **ASCII:** Maps characters to 7-bit binary codes (e.g., 'A' = 01000001).
 - **Unicode:** Supports multiple languages and symbols using 16-bit or 32-bit codes.

File extensions:

.doc, .docx, .pdf, .txt, etc.

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2. Numbers

- **Representation:** Stored in binary format:
 - **Integer Numbers:** Direct binary representation (e.g., 10 = 1010 in binary).
 - **Floating-Point Numbers:** IEEE 754 format with sign, exponent, and mantissa.

3. Images

- **Representation:**
 - Converted into binary by dividing the image into **pixels**.
 - Each pixel is assigned a value representing color, encoded using formats like:
 - **RGB:** Combines red, green, and blue values.
 - **Grayscale:** Represents intensity (single value per pixel).
 - Compressed formats for transmission:
 - **JPEG, PNG** (lossless), **GIF** (animated), **BMP** (uncompressed).

4. Audio

- **Representation:**
 - **Analog to Digital Conversion (ADC):** Sound is sampled at regular intervals.
 - **Encoding Formats:**
 - **MP3, AAC** (lossy compression).
 - **WAV, FLAC** (lossless compression).
 - Audio signal is represented as a sequence of sampled amplitudes.
- **Audio File Formats –**
MP3, M4A audio file type, FLAC, WAV, WMA, AAC, etc.

5. Video

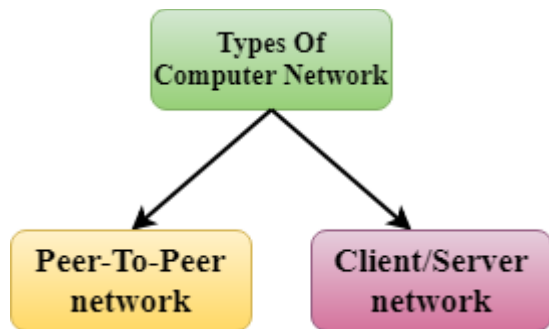
- **Representation:**
 - A sequence of images (frames) displayed over time.
 - Each frame is represented like an image and encoded using video compression standards:
 - **MP4, H.264, AVI** (compressed formats for efficient transmission).
 - Synchronizes audio and video streams for playback.

Video File Formats –
MP4, MOV, AVI, FLV, etc.

Computer Network Architecture:-

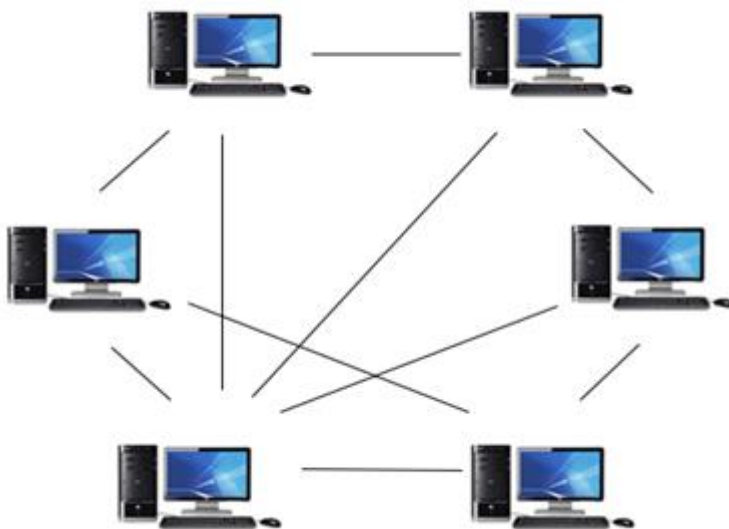
Computer Network Architecture is defined as the physical and logical design of the software, hardware, protocols, and media of the transmission of data. Simply we can say that how computers are organized and how tasks are allocated to the computer.

The two types of network architectures are used:



Peer-To-Peer network

- Peer-To-Peer network is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data.
- Peer-To-Peer network is useful for small environments, usually up to 10 computers.
- Peer-To-Peer network has no dedicated server.

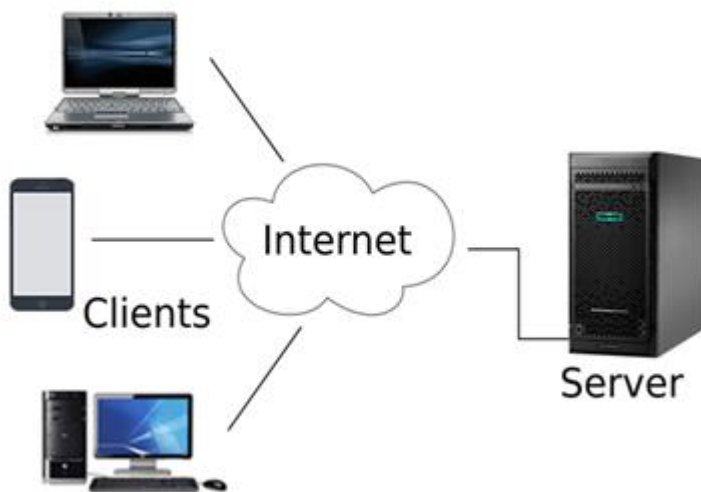


Advantages Of Peer-To-Peer Network:

- It is less costly as it does not contain any dedicated server.
- If one computer stops working but, other computers will not stop working.
- It is easy to set up and maintain as each computer manages itself.

Client/Server Network:-

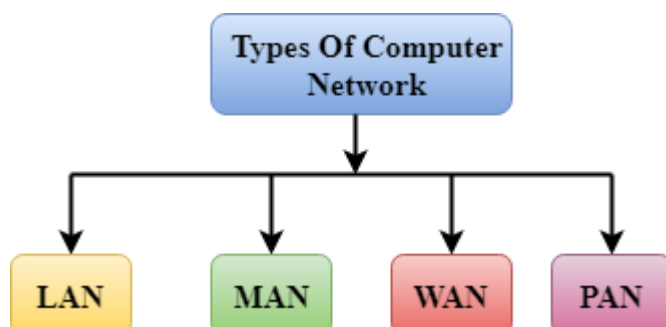
- Client/Server network is a network model designed for the end users called clients, to access the resources such as songs, video, etc. from a central computer known as Server.
- The central controller is known as a **server** while all other computers in the network are called **clients**.
- A server performs all the major operations such as security and network management.
- A server is responsible for managing all the resources such as files, directories, printer, etc.



Advantages Of Client/Server network:

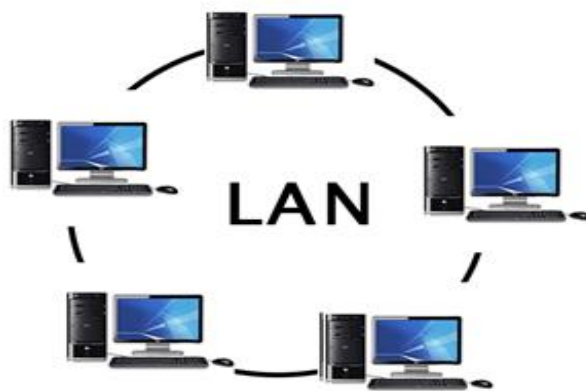
- A Client/Server network contains the centralized system. Therefore we can back up the data easily.
- A Client/Server network has a dedicated server that improves the overall performance of the whole system.
- Security is better in Client/Server network as a single server administers the shared resources.
- It also increases the speed of the sharing resources.

Computer Network Types:-



LAN(Local Area Network)

- Local Area Network is a group of computers connected to each other in a small area such as building, office.
- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.

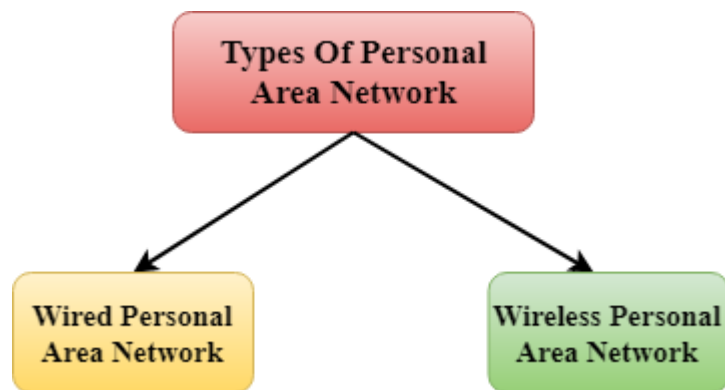


PAN(Personal Area Network)

- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of **30 feet**.



There are two types of Personal Area Network:



MAN(Metropolitan Area Network):-

- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- It has a higher range than Local Area Network(LAN).

Uses Of Metropolitan Area Network:

- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.

WAN(Wide Area Network):-

- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- A Wide Area Network is quite bigger network than the LAN.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.

Internetwork

- An internetwork is defined as two or more computer network LANs or WAN or computer network segments are connected using devices, and they are configured by a local addressing scheme. This process is known as **internetworking**.
- An interconnection between public, private, commercial, industrial, or government computer networks can also be defined as **internetworking**.
- An internetworking uses the **internet protocol**.

Types Of Internetwork:

1. Extranet:-

- An extranet is a communication network based on the internet protocol such as **Transmission Control protocol** and **internet protocol**.
- It is used for information sharing.
- An extranet is the lowest level of internetworking. It can be categorized as **MAN, WAN** or other computer networks.

2. Intranet:-

- An intranet is a private network based on the internet protocol such as **Transmission Control protocol** and **internet protocol**
- The main aim of the intranet is to share the information and resources among the organization employees.
- An intranet provides the facility to work in groups and for teleconferences.

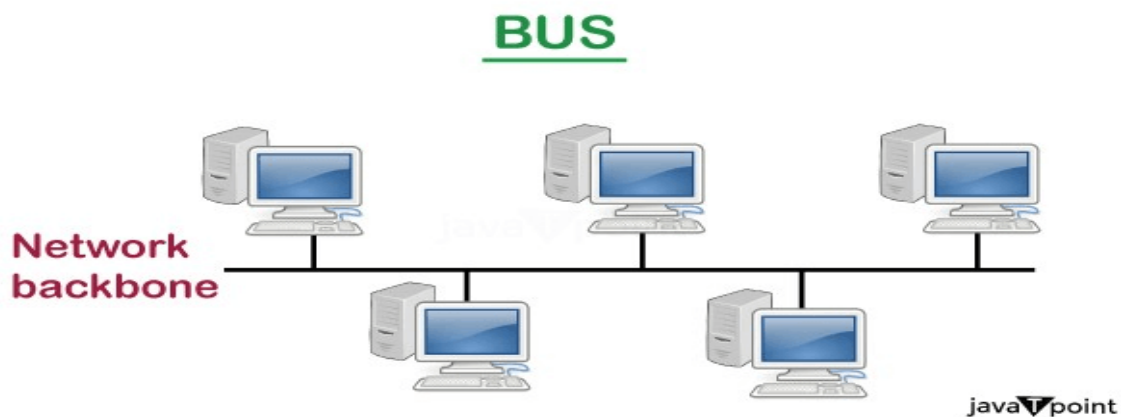
What is Network Topology?

Topology defines the structure of the network of how all the components are interconnected to each other.

Some of the Network Topologies are:-

1. BUS 2. RING 3. STAR 4. MESH 5. TREE 6. HYBRID

1) Bus Topology



- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- The bus topology is mainly used in 802.3 (ethernet) and 802.4 standard networks.
- The configuration of a bus topology is quite simpler as compared to other topologies.

Advantages:

- Easy and inexpensive to set up.
- Requires less cable compared to other topologies.

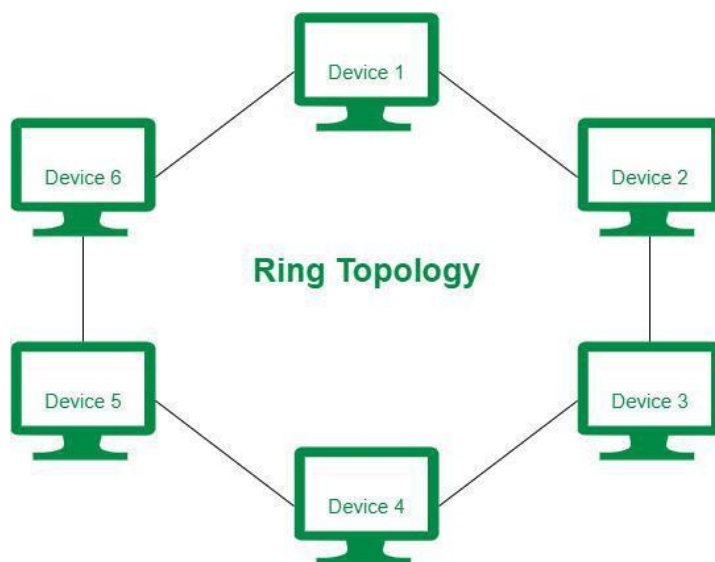
- Suitable for small networks or temporary setups.

Disadvantages:

- Single point of failure (if the central cable fails, the entire network goes down).
- Troubleshooting can be complex.
- Performance degrades with increased devices and traffic.
- Limited scalability.

2)Ring Topology

- Ring topology is like a bus topology, but with connected ends.
- The data flows in one direction, i.e., it is unidirectional.
- The data in a ring topology flow in a clockwise direction.

**Advantages of Ring Topology**

- The data transmission is high-speed.
- The possibility of collision is minimum in this type of topology.
- Cheap to install and expand.
- It is less costly than a star topology.

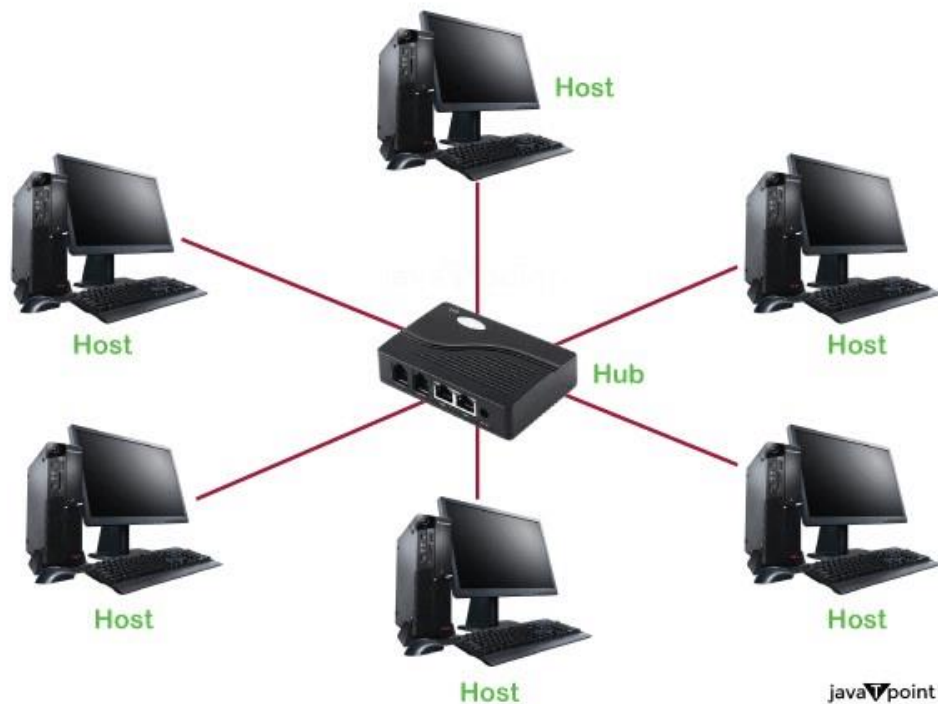
Disadvantages of Ring Topology

- The failure of a single node in the network can cause the entire network to fail.
- Troubleshooting is difficult in this topology.
- The addition of stations in between or the removal of stations can disturb the whole topology.
- Less secure.

3) Star Topology:-

Star Topology

- All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection.
- Point-to-point connection between hosts and hub.



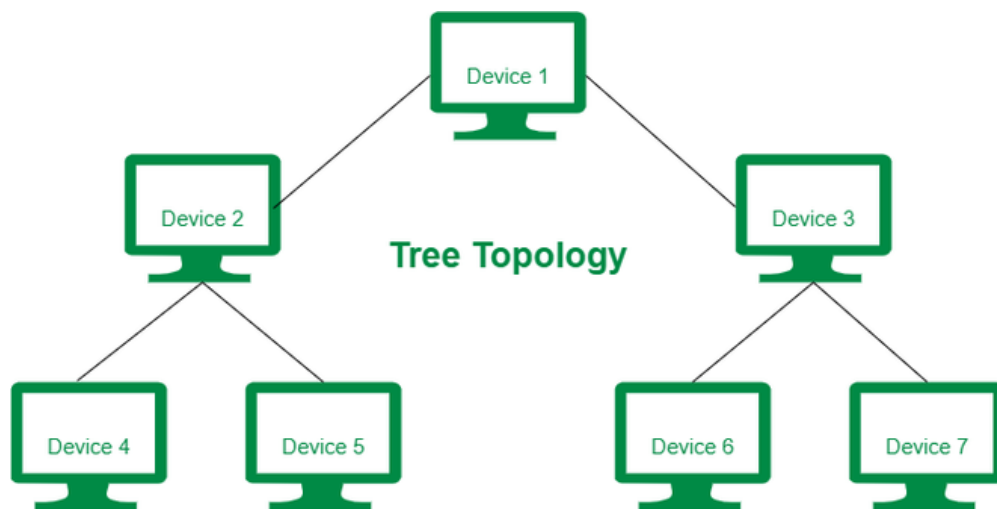
Advantages:

- Easy to install and manage.
- Centralized control for troubleshooting.
- If one device fails, it doesn't affect the rest of the network.
- Scalable and flexible for expansion.

Disadvantages:

- Expensive due to the need for a central hub or switch.
- If the central hub fails, the entire network is affected.
- Requires more cabling than bus topology.

4) Tree topology:-



- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion.

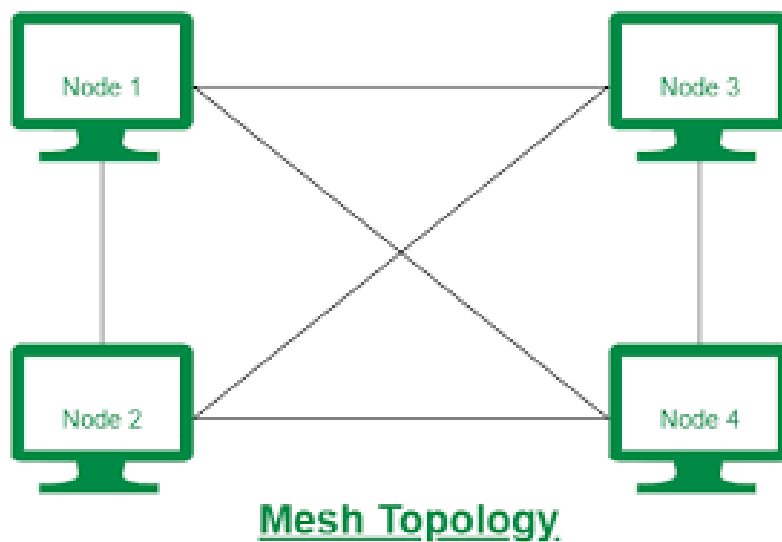
Advantages:

- Scalable and easy to expand.
- Combines the benefits of bus and star topologies.
- Centralized management, making it easier to control traffic.
- Isolation of segments improves performance.

Disadvantages:

- Complex structure, requiring more cabling.
- If the main backbone fails, large parts of the network are affected.
- Expensive and time-consuming to set up and maintain.
- Can be prone to congestion at the root level.

5) Mesh topology:-



- Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- Mesh topology is mainly used for wireless networks.
- Mesh topology can be formed by using the formula:
Number of cables = $(n*(n-1))/2$;

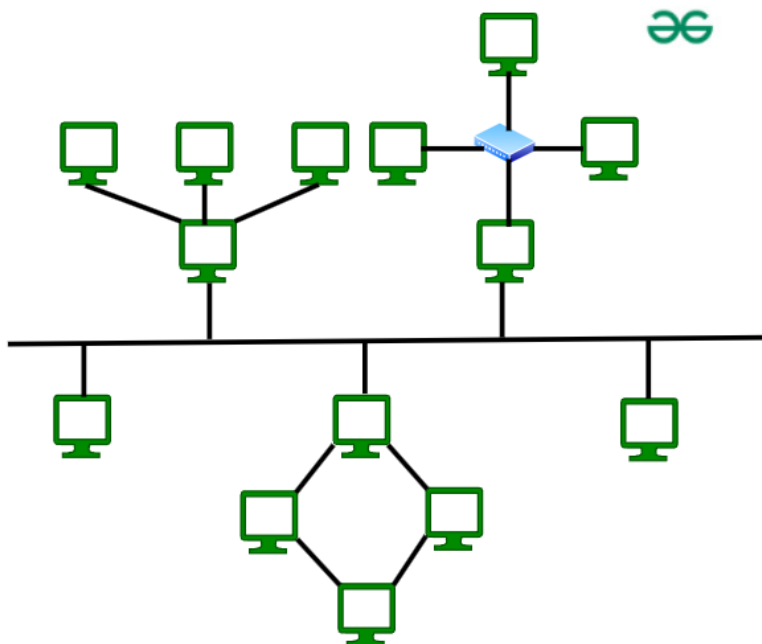
Advantages:

- High reliability due to multiple paths.
- Fault-tolerant (a failed connection doesn't affect the network).
- Supports high data traffic and reduces congestion.

Disadvantages:

- Expensive due to extensive cabling.
- Complex installation and maintenance.
- Requires more hardware, increasing costs.

6) Hybrid Topology:-



- The combination of various different topologies is known as Hybrid topology.
- A Hybrid topology is a connection between different links and nodes to transfer the data.

Advantages:

- Combines strengths of different topologies.
- Scalable and flexible for expansion.
- Reliable due to isolated faults in individual segments.

Disadvantages:

- Expensive to design and implement.
- Complex setup and maintenance.
- Requires skilled administration.

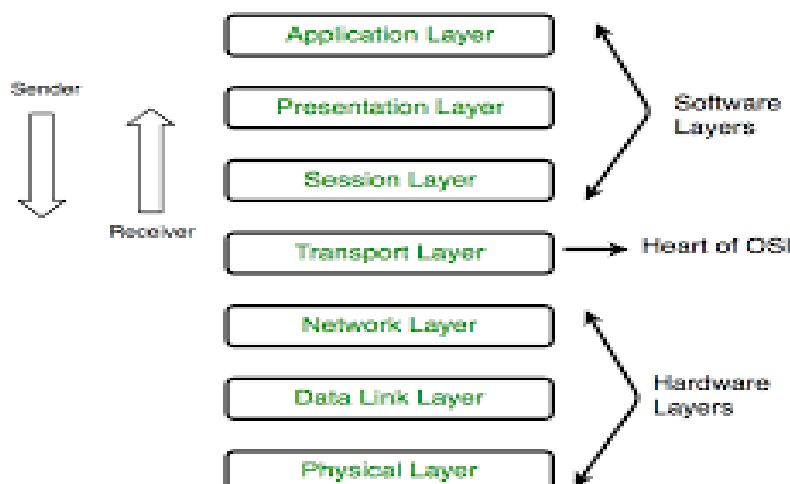
What is OSI Model?

- The OSI (Open Systems Interconnection) Model is a set of rules that explains how different computer systems communicate over a network.
- OSI Model was developed by the International Organization for Standardization (ISO) in 1984.
- The OSI Model consists of 7 layers and each layer has specific functions and responsibilities.

7 Layers of OSI Model

There are the seven OSI layers. Each layer has different functions. A list of seven layers are given below:

1. Physical Layer
2. Data-Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer



OSI Model Layers:

1. Physical Layer:

- The lowest layer of the OSI reference model is the Physical Layer.
- It is responsible for the actual physical connection between the devices.
- The physical layer contains information in the form of bits.
- Common physical layer devices are [Hub](#), [Repeater](#), [Modem](#), and [Cables](#).

2. Data Link Layer:

- The data link layer is responsible for the node-to-node delivery of the message.
- The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer.

The Data Link Layer is divided into two sublayers:

[Logical Link Control \(LLC\)](#)

[Media Access Control \(MAC\)](#)

3. Network Layer:

- It is a layer 3 that manages device addressing, tracks the location of devices on the network.
 - The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.
-

4. Transport Layer:

- Ensures reliable data delivery with error checking and retransmission.
- The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.
- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.

5. Session Layer:

- Establishes, maintains, and terminates communication sessions.
- Session Layer in the OSI Model is responsible for the establishment of connections, management of connections, terminations of sessions between two devices.
- It also provides authentication and security.
- Protocols used in the Session Layer are NetBIOS, PPTP.

6. Presentation Layer:

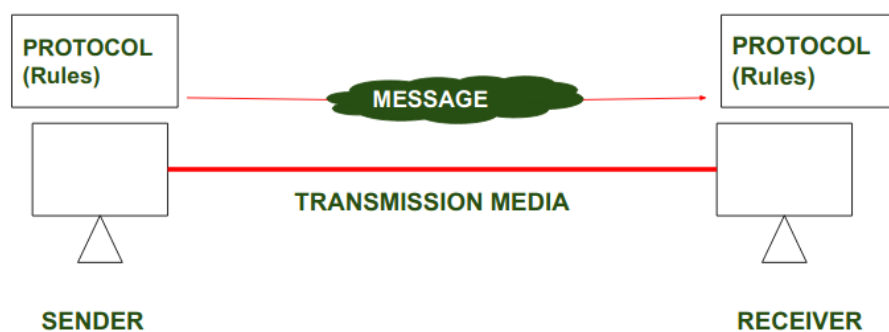
- Translates, encrypts, or compresses data for the application layer.
 - A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
 - It acts as a data translator for a network.
 - The Presentation layer is also known as the syntax layer.
-

7. Application Layer:

- An application layer serves as a window for users and application processes to access network service.
- It handles issues such as network transparency, resource allocation, etc. This layer provides the network services to the end-users.
- An application layer serves as a window for users and application processes to access network service.

What is Protocol?

- A protocol is a set of rules that determines how data is sent and received over a network.
- The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly.
- Protocols help make sure that data moves smoothly and securely between devices on a network.



Protocols

Types of protocols:

1. **Network Layer Protocols:** Operate in Layer 3 for routing, forwarding, and addressing. Examples: IP, ICMP.
2. **Transport Layer Protocols:** Provide end-to-end data transfer between apps. Examples: TCP, UDP.
3. **Application Layer Protocols:** Enable communication between applications, handling data format and interpretation. Examples: HTTP, FTP, SMTP.
4. **Wireless Protocols:** Support data transfer over wireless networks. Examples: Bluetooth, Wi-Fi, LTE.
5. **Routing Protocols:** Determine the best paths for data transmission. Examples: RIP, OSPF, BGP.
6. **Security Protocols:** Ensure data confidentiality, integrity, and authenticity. Examples: SSL, TLS.
7. **Internet Protocols:** Handle device identification and data routing with unique addressing. Example: IP.

Key protocols in computer networks:

1. **TCP:** Ensures accurate data transmission.
2. **IP:** Routes packets to destinations.
3. **HTTP/HTTPS:** Transfers web pages; HTTPS adds encryption.
4. **FTP:** Transfers files between devices.
5. **SMTP:** Sends emails.

What is Standards?

- Standards are the set of rules for data communication that are needed for the exchange of information among devices.
- It is important to follow Standards which are created by various Standard Organizations like IEEE, ISO, ANSI, etc.

Types of Standards:

1. **De Facto:** Informal standards adopted due to widespread use, often set by manufacturers (e.g., Apple's and Google's proprietary rules).
2. **De Jure:** Formal standards officially approved by organizations like ANSI, IEEE, or ISO (e.g., TCP, IP, SMTP).

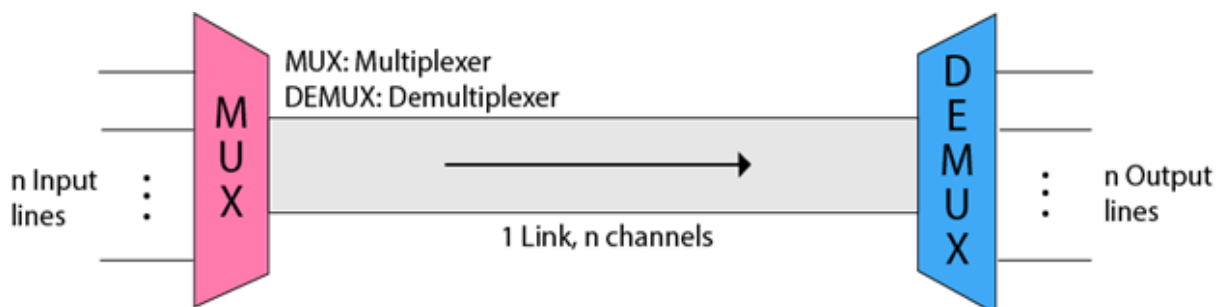
What is Multiplexing?

Multiplexing is a technique used to combine and send the multiple data streams over a single medium.

The process of combining the data streams is known as multiplexing and hardware used for multiplexing is known as a multiplexer.

Multiplexing is achieved by using a device called Multiplexer (**MUX**) that combines n input lines to generate a single output line. Multiplexing follows many-to-one, i.e., n input lines and one output line.

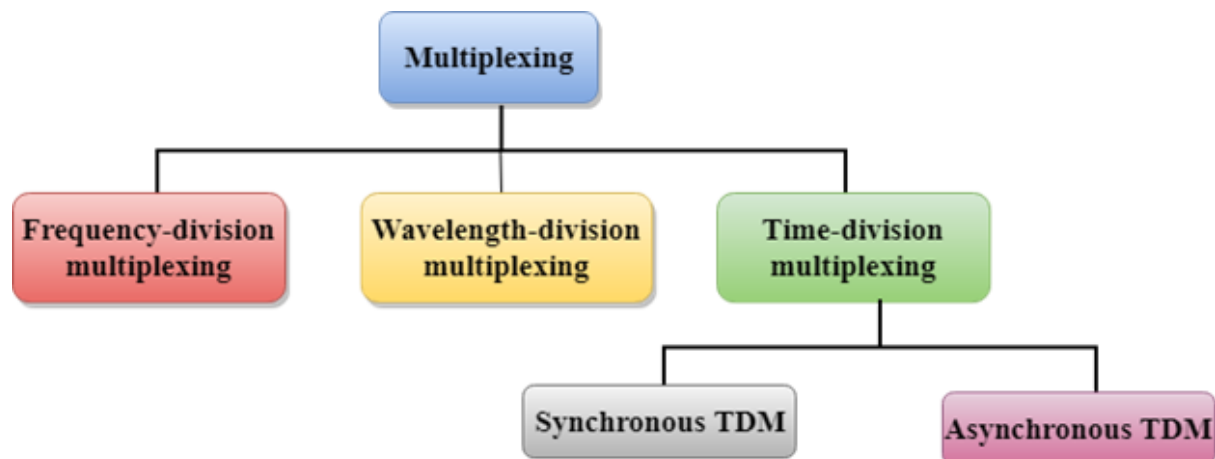
Demultiplexing is achieved by using a device called Demultiplexer (**DEMUX**) available at the receiving end. DEMUX separates a signal into its component signals (one input and n outputs).



Advantages of Multiplexing:

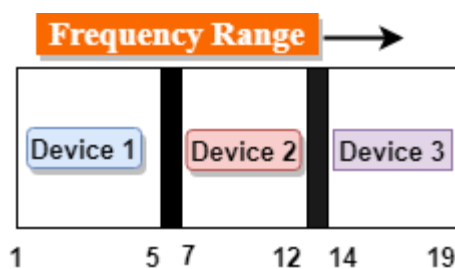
- More than one signal can be sent over a single medium.
- The bandwidth of a medium can be utilized effectively.

Multiplexing Techniques:- Multiplexing techniques can be classified as:



1. Frequency-division Multiplexing (FDM)

- It is an analog technique.
- Frequency Division Multiplexing is a technique in which the available bandwidth of a single transmission medium is subdivided into several channels.



- The main aim of the FDM is to subdivide the available bandwidth into different frequency channels and allocate them to different devices.
- **FDM** is mainly used in radio broadcasts and TV networks.

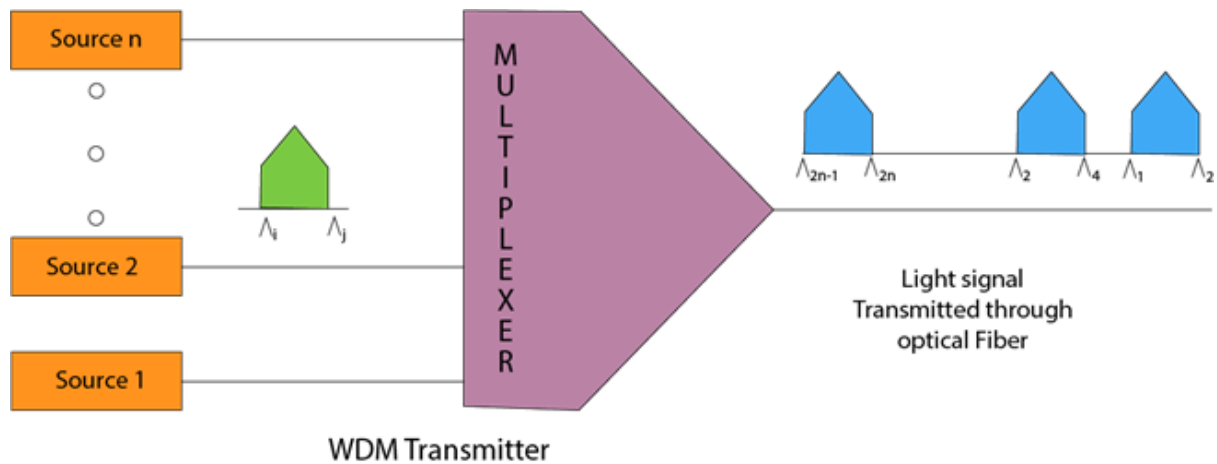
Advantages Of FDM:

- FDM is used for analog signals.
- FDM process is very simple and easy modulation.
- A Large number of signals can be sent through an FDM simultaneously.
- It does not require any synchronization between sender and receiver.

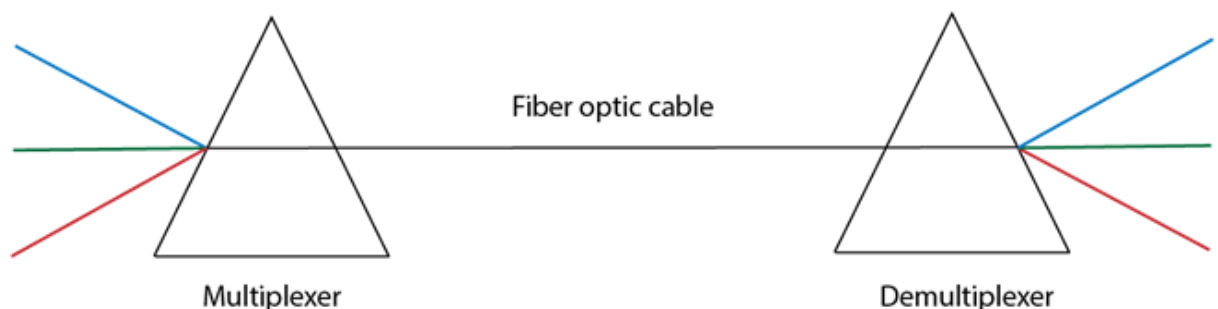
Disadvantages Of FDM:

- FDM technique is used only when low-speed channels are required.
- It suffers the problem of crosstalk.
- A Large number of modulators are required.
- It requires a high bandwidth channel.

2.Wavelength Division Multiplexing (WDM)



- Wavelength Division Multiplexing is same as FDM except that the optical signals are transmitted through the fibre optic cable.
- WDM is used on fibre optics to increase the capacity of a single fibre.
- It is used to utilize the high data rate capability of fibre optic cable.
- It is an analog multiplexing technique.
- Multiplexing and Demultiplexing can be achieved by using a prism.
- Prism also performs a reverse operation, i.e., demultiplexing the signal.



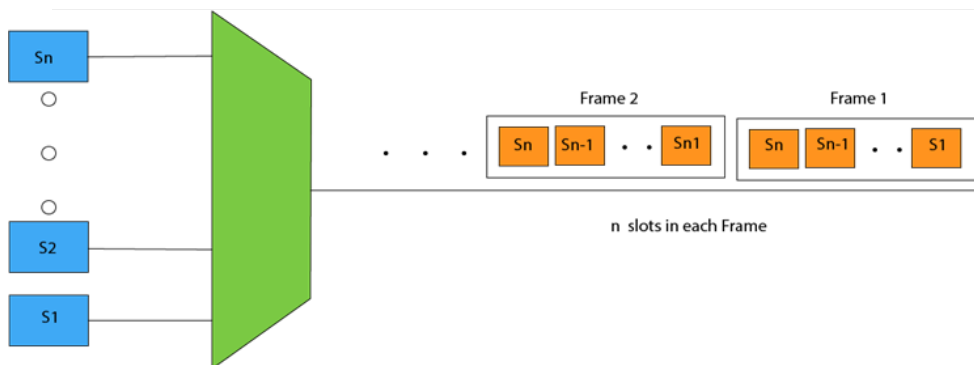
3. Time Division Multiplexing

- It is a digital technique.
- In TDM, the signal is transmitted in the form of frames. Frames contain a cycle of time slots in which each frame contains one or more slots dedicated to each user.
- It can be used to multiplex both digital and analog signals but mainly used to multiplex digital signals.

There are two types of TDM:

A. Synchronous TDM

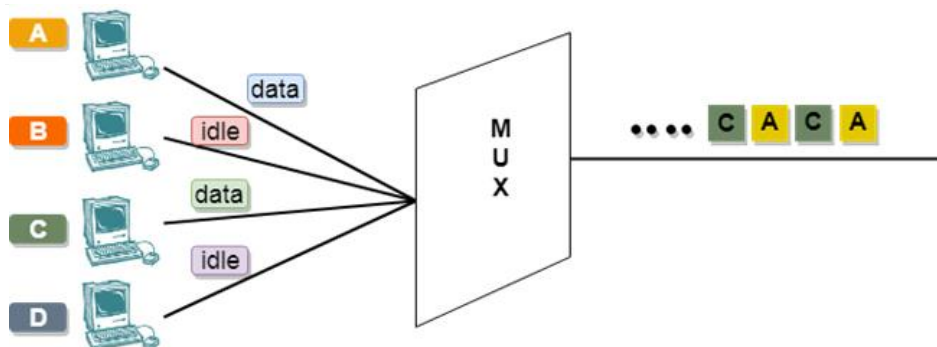
- A Synchronous TDM is a technique in which time slot is preassigned to every device.
- In Synchronous TDM, each device is given some time slot irrespective of the fact that the device contains the data or not.
- If the device does not have any data, then the slot will remain empty.
- If there are n devices, then there are n slots.



Concept Of Synchronous TDM

B. Asynchronous TDM

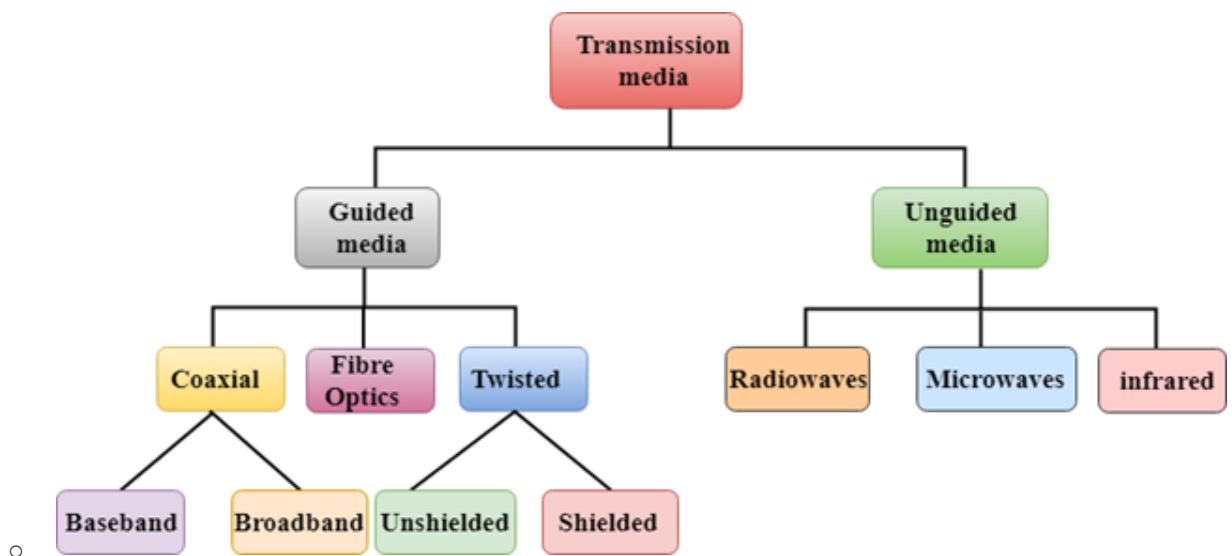
- An asynchronous TDM technique dynamically allocates the time slots to the devices.
- An asynchronous TDM is also known as Statistical TDM.
- In Asynchronous TDM, total speed of the input lines can be greater than the capacity of the channel.



What is Transmission media?

- Transmission media is a communication channel that carries the information from the sender to the receiver.
- Data is transmitted through the electromagnetic signals.
- The transmission media is available in the lowest layer of the OSI reference model, i.e., **Physical layer**.

Classification Of Transmission Media:

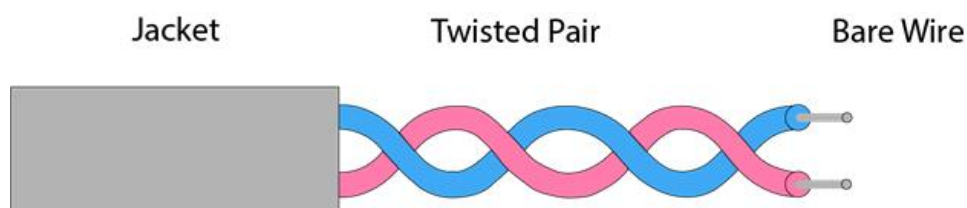


1. Guided Media

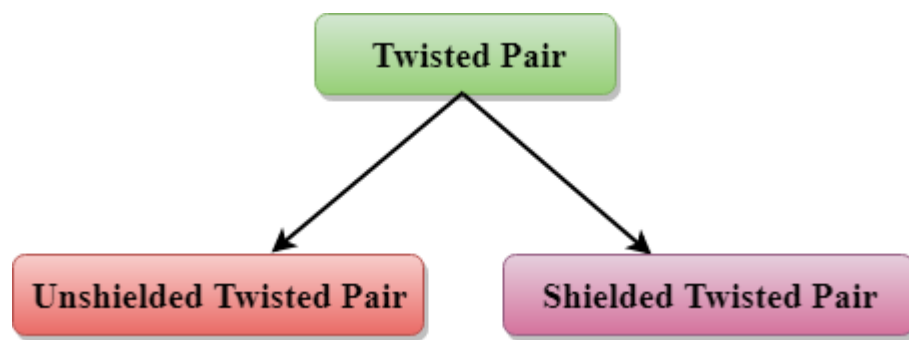
Guided Media is also referred to as Wired or Bounded transmission media. Signals being transmitted are directed and confined in a narrow pathway by using physical links.

A. Twisted Pair Cable:

- Twisted pair is a physical media made up of a pair of cables twisted with each other.
- A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable.
- The frequency range for twisted pair cable is from 0 to 3.5KHz.



Types of Twisted pair:



Unshielded Twisted Pair:

- An unshielded twisted pair is widely used in telecommunication
- [UTP](#) consists of two insulated copper wires twisted around one another.

Advantages of Unshielded Twisted Pair

- Least expensive
- Easy to install
- High-speed capacity

Disadvantages of Unshielded Twisted Pair

- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

Shielded Twisted Pair:

[Shielded Twisted Pair \(STP\)](#) cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference.

It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages of Shielded Twisted Pair

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk

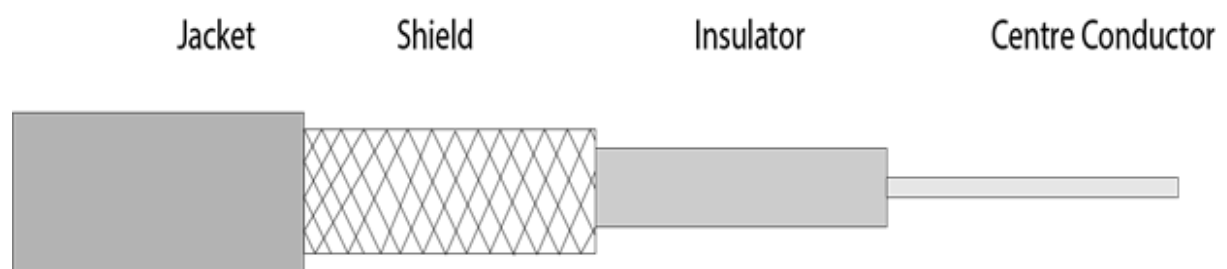
- Comparatively faster

Disadvantages of Shielded Twisted Pair

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

B. Coaxial Cable

- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.



Advantages Of Coaxial cable:

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages Of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

Coaxial cable is of two types:

1. **Baseband transmission:** It is defined as the process of transmitting a single signal at high speed.

- 2. Broadband transmission:** It is defined as the process of transmitting multiple signals simultaneously.
-

C. Fibre Optic

- Fibre optic cable is a cable that uses electrical signals for communication.
 - Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
 - The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
 - Fibre optics provide faster data transmission than copper wires.
-

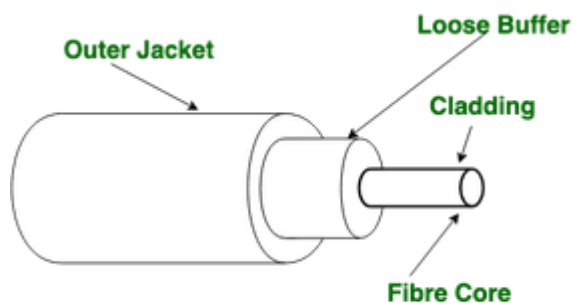


Figure of Optical Fibre Cable

Advantages of Optical Fibre Cable

- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages of Optical Fibre Cable

- Difficult to install and maintain
- High cost

2. UnGuided Transmission

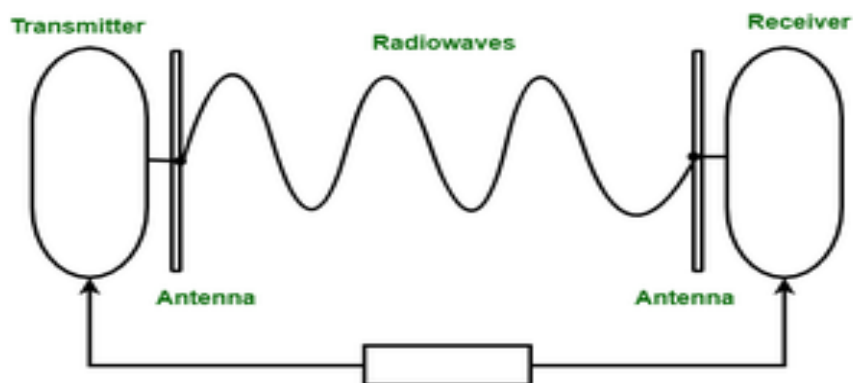
- An unguided transmission transmits the electromagnetic waves without using any physical medium. Therefore it is also known as wireless transmission.
- In unguided media, air is the media through which the electromagnetic energy can flow easily.
- No physical medium is required for the transmission of electromagnetic signals.

Features of Unguided Media

- The signal is broadcasted through air
- Less Secure
- Used for larger distances

A. Radio Waves

- [Radio waves](#) are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned.
- Frequency Range: 3KHz – 1GHz. AM and FM radios and cordless phones use Radio waves for transmission.



B. Microwaves

- It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other.
- The distance covered by the signal is directly proportional to the height of the antenna. Frequency Range: 1GHz – 300GHz.
- [Micro waves](#) are majorly used for mobile phone communication and television distribution.

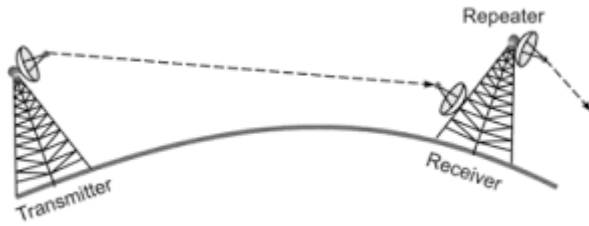


Fig: Microwave Transmission

C. Infrared

- [Infrared waves](#) are used for very short distance communication.
- They cannot penetrate through obstacles. This prevents interference between systems.
Frequency Range: 300GHz – 400THz.
- It is used in TV remotes, wireless mouse, keyboard, printer, etc.



Television



Infrared Radiations



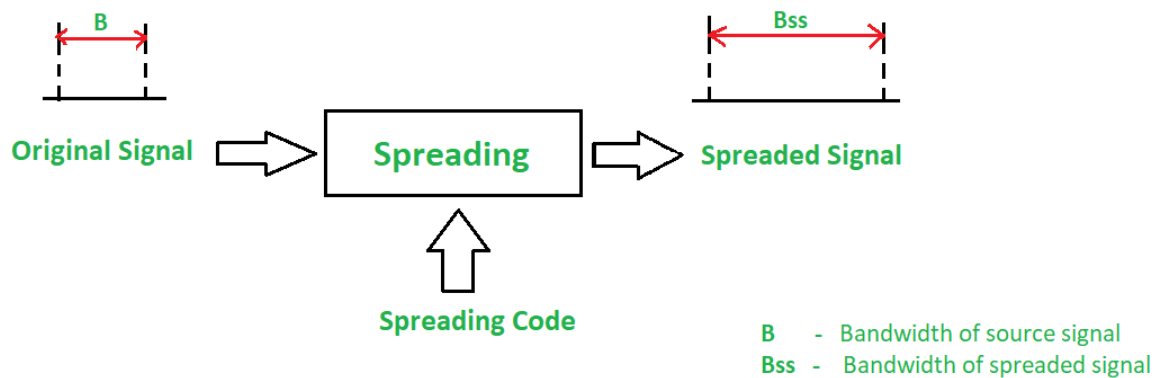
Remote

Transmission Media	Application
Unshielded Twisted Pair (UTP)	Local Area Networks (LAN), telephones
Shielded Twisted Pair (STP)	Industrial networks, environments with high interference
Optical Fiber Cable	Long-distance communication, internet backbones
Coaxial Cable	Cable TV, broadband internet, CCTV
Stripline	Printed Circuit Boards (PCBs), microwave circuits
Microstripline	Antennas, satellite communication, RF circuits

Transmission Media	Application
Radio	Wireless communication, AM/FM radio, mobile phones
Infrared	Remote controls, short-range communication
Microwave	Satellite communication, radar, long-distance links

What is Spread Spectrum?

- Spread spectrum is a method of transmitting [radio signals](#) over a wide range of frequencies.
- Spread Spectrum is a wireless communication technology that distributes the transmitted signal across a larger bandwidth than the original signal



Spread Spectrum